# Samardip Mandal

**Frontend Engineer** | React · Next.js · TypeScript | Embeddable SDKs, Real-time AI Interfaces & 3D

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## Summary

Frontend engineer with 4 years of experience in React, Next.js and TypeScript; currently Member of Technical Staff at Styldod, promoted from SDE I. Built the frontend of REimagineHome's multi-tenant, embeddable AI widget SDK (**3M+ AI generations**), used by real-estate portals including Otodom, and ship core features of **REimagineHome**, an AI interior-design app with **2M+ users**, from real-time AI streaming interfaces to Three.js 3D room views.

## Skills

**Languages & Frameworks:** TypeScript, JavaScript (ES6+), React.js, Next.js, Preact, HTML5, CSS3
**State & Data:** Redux Toolkit, React Query (TanStack Query), RxJS, REST, GraphQL (Apollo Client), WebSockets
**UI & Graphics:** Material UI, Emotion, SCSS, Tailwind CSS, Three.js, responsive and mobile-first UI
**SDK & Platform:** iframe SDKs, postMessage, multi-tenant white-label config, i18n (next-intl), npm workspaces
**Build & Production:** Vite, esbuild/tsup, Rollup, Sentry, PostHog, Google Tag Manager, AWS Cognito, Paddle, Vercel, Git

## Experience

**Styldod Inc** | Bengaluru **Jan 2025 – Present**
**Member of Technical Staff – Frontend** *(promoted from Software Development Engineer I)*

- **Embed SDK:** Built the frontend of REimagineHome Embed, a white-label AI virtual-staging widget that partners add with one script tag: a **∼9 KB gzipped** loader (tsup/esbuild) that opens a Preact + Vite iframe app.
- Designed the **multi-tenant UI architecture**: one shared logic layer with per-tenant UI modules (default, Otodom) lazy-loaded by org slug; branding, fonts, sections and feature flags (uploads, history, analytics) merge from org config and host overrides, so partners are configured, not forked.
- Wrote the typed host–iframe **postMessage protocol** (init, theme, close, destroy, CTA actions, errors) with origin and source checks on both sides, plus publishable-key sessions limited to allowlisted domains.
- Shipped **i18n in 5 locales** (en-US, en-GB, de, es-ES, pl-PL) loaded on demand, real-time generation status over WebSockets, and event analytics sent with `fetch` keepalive so events survive widget teardown.
- **REimagineHome 2.0** (Next.js, Redux Toolkit, React Query, RxJS), the core app with **2M+ users**: built the streaming "thinking" interface that shows the AI agent's reasoning step by step and persists across reloads, the chat-driven design flow with shoppable product bundles, and live WebSocket reconnection status.
- Built the AI photo **quality-check flow**: quality report, per-photo flags, and a batch "Fix all" pipeline with polling, credit checks and processing states; shipped new pricing tiers on Paddle with Cloudflare Turnstile.
- Improved production reliability: filtered browser noise in Sentry with `beforeSend` and fixed the surfaced errors, recovered HEIC photos mislabelled as JPEG, and added polyfills for older browsers.
- **Photos-to-3D** (Next.js, TypeScript, Three.js): set up the app, Cognito sign-in, the room-to-3D job pipeline (API client, Redux slice, WebSocket routing) and a dollhouse camera that fades walls blocking the view.
- **DormTok** (Next.js, React Query, Redux), a social platform for AI room-makeover videos: bootstrapped the app and built upload, credits and WebSocket integration, Cognito route guards with token refresh, a registry flow with Amazon and Walmart checkout, and an internal analytics dashboard.
- **Smart Media Module** (Next.js, Redux, GraphQL): built the listing-photo compliance editor with a draw/erase mask toolbar, generation-workflow state management, and a long-polling fallback for WebSocket media updates.
- Lead frontend architecture and code reviews across these products, and mentor engineers on the team.

**Global IDs** | Kolkata **Jun 2023 – Dec 2024**
**Software Developer**

- Built responsive production web applications and complex UI workflows in React.js, JavaScript, HTML5 and CSS3.
- Refactored legacy frontend modules into reusable React components, removing duplicated UI code and fixing rendering bottlenecks through better component structure and state management.

**Tale Wallet** | Bengaluru **Sep 2022 – May 2023**
**Associate Software Engineer**

- Built the React frontend of a cryptocurrency wallet: reusable components, application state management and backend API integration, with real-time rendering of wallet data.

## Education

**B.Tech, Computer Science and Engineering** | CGPA 8.98/10 **Aug 2018 – Jun 2022**
RCC Institute of Information Technology, Kolkata

## Achievements

**CodeChef:** 4⋆ (max rating 1803) | **Google Code Jam 2021:** Qualified